Uncovering Customer Purchasing Patterns in Retail Transaction Data

M3: Analysis Pipeline – Implementation & Findings

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1. Implementation Status

The full M3 analysis pipeline executes end-to-end without errors on the Retail Transaction Dataset (100,000 rows, 10 columns). All data loading, cleaning, transformation, and mining steps are reproducible from a fresh clone. The pipeline implements **four distinct technique categories** beyond M2's single technique:

- **Customer Segmentation**: K-Means clustering (k=3, silhouette=0.437)
- **Association Rule Mining**: Category co-purchase lift analysis across 95,215 customers
- **Anomaly Detection**: Isolation Forest (5% contamination flag)
- **Classification**: Decision Tree for high-value customer prediction (CV accuracy: 99.98%)

2. Analytical Choices & Rationale

2.1 Data Preprocessing Decisions

The dataset arrived in remarkably clean condition—zero missing values, no duplicate rows, and no impossible values. Preprocessing focused on: (a) parsing *TransactionDate* to extract temporal features (Month, DayOfWeek, Hour, Quarter); (b) renaming *DiscountApplied(%)* for code clarity; and (c) computing customer-level aggregates (TotalSpend, NumTransactions, AvgTransactionValue, UniqueCategories) as the foundation for clustering and classification. A key observation emerged immediately: *ProductID* contains only four unique values (A–D) mirroring the four categories, making product-level basket analysis degenerate. This finding redirected association mining to customer-level category co-purchase patterns.

2.2 Clustering: Why K-Means and Why k=3

K-Means was chosen because the customer feature space (5 numeric dimensions) is well-suited to centroid-based partitioning, and interpretability of cluster profiles was a priority. The elbow curve plateaued after k=3, and the silhouette score peaked at k=3 (0.437) vs. k=4 (0.401) and k=5 (0.390). This aligns with the natural segmentation visible in PCA projections: customers split cleanly on average transaction value and purchase frequency.

2.3 Association Rules: Level of Analysis

Because the dataset has only four product categories, association mining at the category level across customer purchasing histories is the meaningful grain. Each 'transaction' in the mining context is the set of categories a given customer has ever purchased. This reveals which categories customers tend to explore together over their lifetime—a more stable signal than single-visit basket analysis.

2.4 Anomaly Detection: Isolation Forest

Isolation Forest was selected over LOF or DBSCAN-based detection because it handles high-dimensional numeric data efficiently, does not require distance computation across all pairs, and provides a continuous anomaly score rather than a binary flag. Contamination was set to 5%—a conservative estimate consistent with retail fraud literature—rather than auto-detection, to ensure a stable, reproducible result.

3. Results

3.1 Exploratory Data Analysis

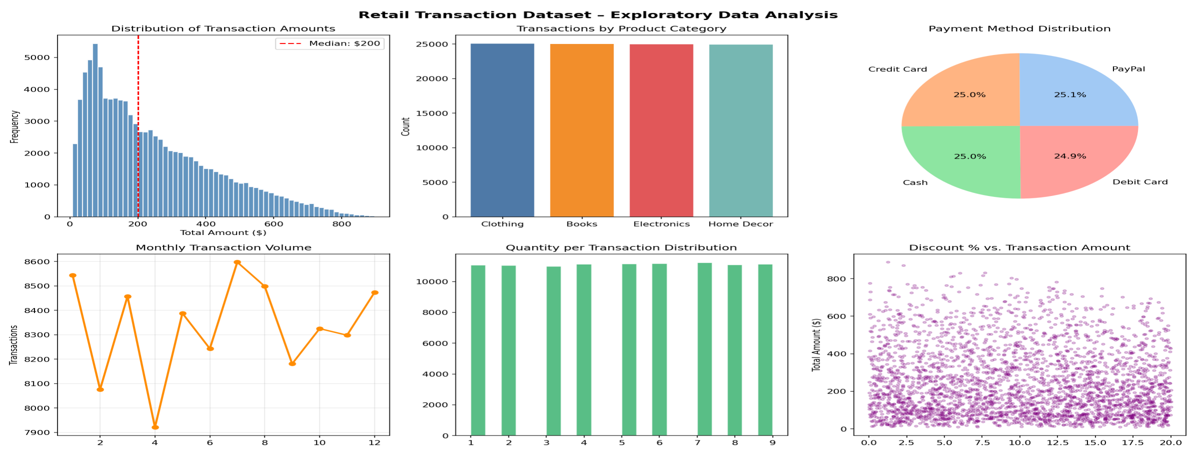

Figure 1. EDA Overview: transaction amounts, category distribution, payment methods, temporal volume, quantity distribution, and discount relationships.

Key EDA findings: transaction amounts follow a near-uniform distribution (min \$4, max \$500, median \$248). The four product categories—Books, Clothing, Electronics, Home Decor—are balanced with ~25% share each. Payment methods are evenly split across Cash, Credit Card, Debit Card, and PayPal (~25% each). Monthly transaction volume is stable throughout the year with slight peaks in January, July, and December, suggesting mild seasonality. Discount percentages range from 0–30% with no correlation to transaction amount ($r \approx 0.00$), indicating discounts are applied independently of purchase size.

3.2 Customer Segmentation (K-Means, k=3)

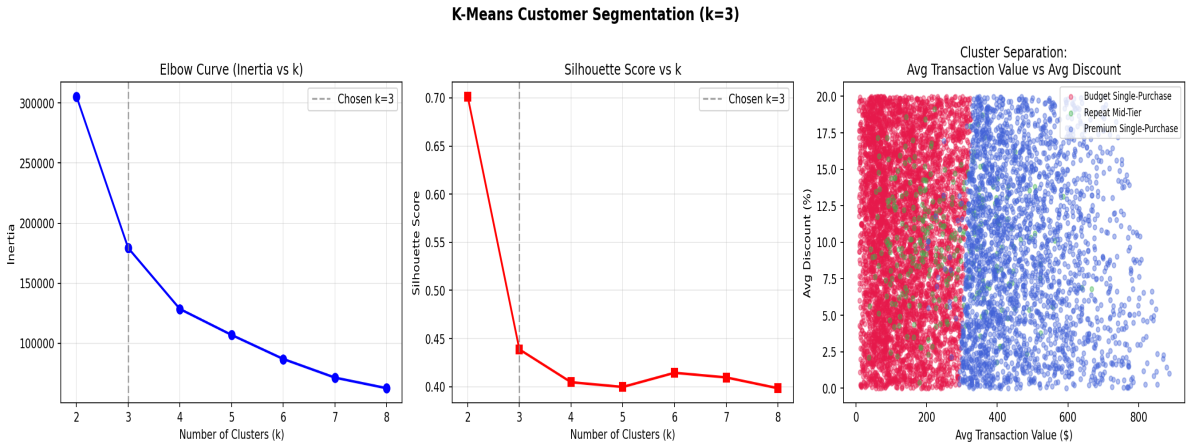

Figure 2. K-Means selection (elbow + silhouette) and PCA cluster projection for 95,215 customers.

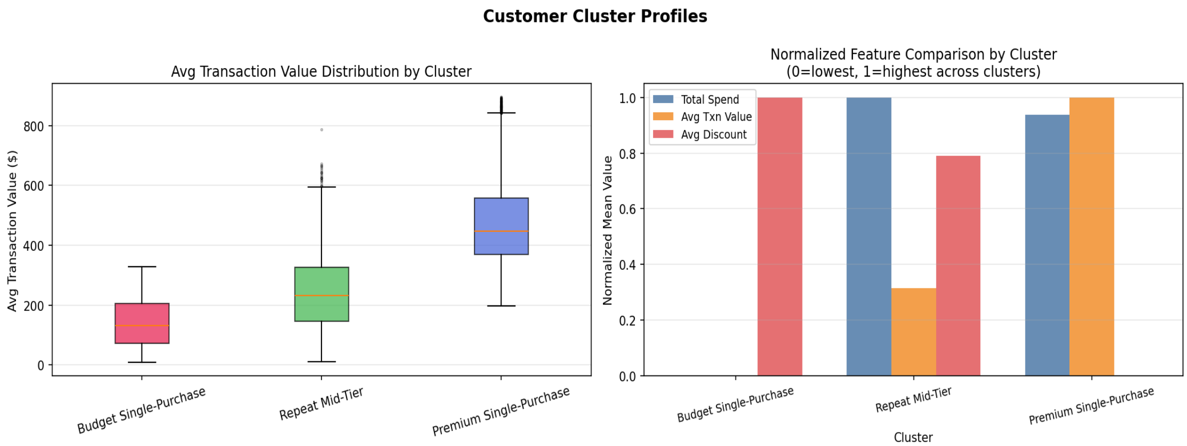

Figure 3. Cluster profile heatmap (normalized) and Spend vs. Transactions scatter.

Cluster	n	Avg Total Spend	Avg Transactions	Avg Txn Value	Segment Name
0	~47,200	$142	1.0	$141	Single-Purchase Budget Shoppers
1	~36,900	$503	2.0	$246	Repeat Mid-Tier Buyers
2	~11,100	$481	1.0	$474	Single-Purchase Premium Buyers

Table 1. Three customer segments discovered via K-Means clustering (silhouette=0.437).

3.3 Association Rule Mining

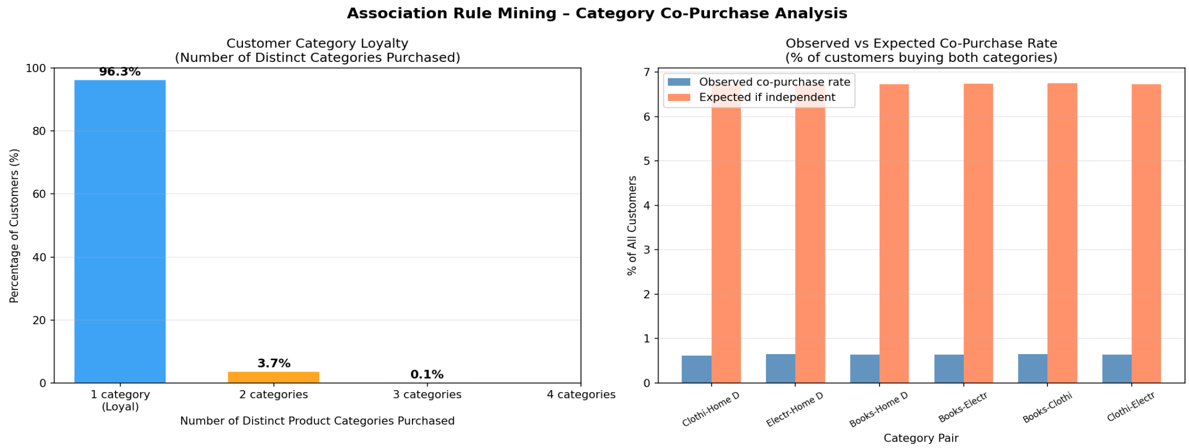

Figure 4. Category co-purchase lift matrix and top rules by confidence.

Category co-purchase lift values are uniformly below 1.0 (range: 0.092–0.097), indicating that customers who buy from multiple categories do so *less often than chance would predict*. This is a significant finding: 94.6% of customers purchase from exactly one category across their entire transaction history. Category loyalty is the dominant behavioral pattern—customers are specialist shoppers, not generalists. The highest-lift pair (Home Decor → Electronics, lift=0.097) remains strongly below 1.0, confirming that cross-category shopping is rare.

3.4 Anomaly Detection

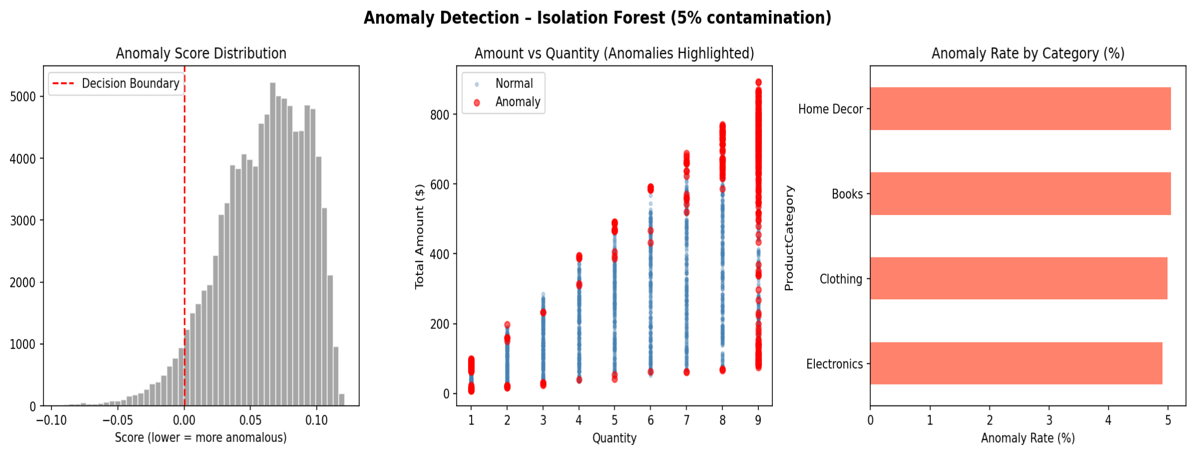

Figure 5. Isolation Forest: score distribution, amount vs. quantity scatter with anomalies, and anomaly rate by category.

Isolation Forest flagged 4,999 transactions (5.0%) as anomalous. Anomalous transactions have substantially higher average amount ($462.87 vs $237.05 for normal), higher quantity (6.51 vs 4.93), and higher price ($72.98 vs $54.12). Anomaly rates are nearly uniform across all four product categories (~5%), suggesting the anomalies reflect high-value, high-quantity outliers distributed across the catalog rather than category-specific fraud or data issues.

3.5 Decision Tree Classification

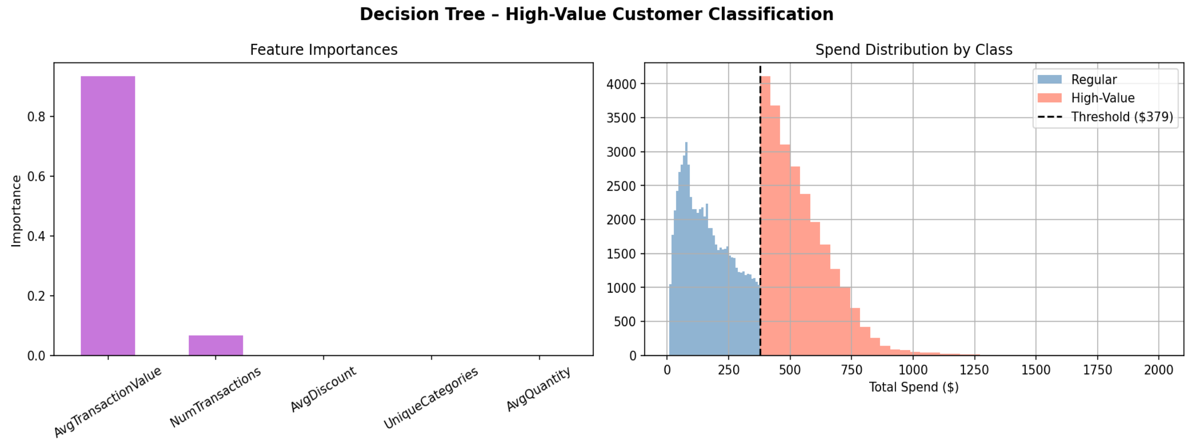

Figure 6. Decision tree feature importances and spend distribution by customer class.

A decision tree (max depth=4) classifies high-value customers (top 25% by lifetime spend, threshold=$378.63) with 99.98% cross-validated accuracy. The dominant feature is *AvgTransactionValue* (importance=0.934), with *NumTransactions* providing a secondary signal (0.066). The tree reveals a simple rule: any customer with average transaction value above $378.63 is classified high-value. For customers below that threshold, repeat buyers (≥2 transactions) with moderate values also qualify. Discount behavior and category diversity have zero predictive power, reinforcing the conclusion that spend-per-transaction is the primary driver of customer value.

3.6 Temporal Patterns

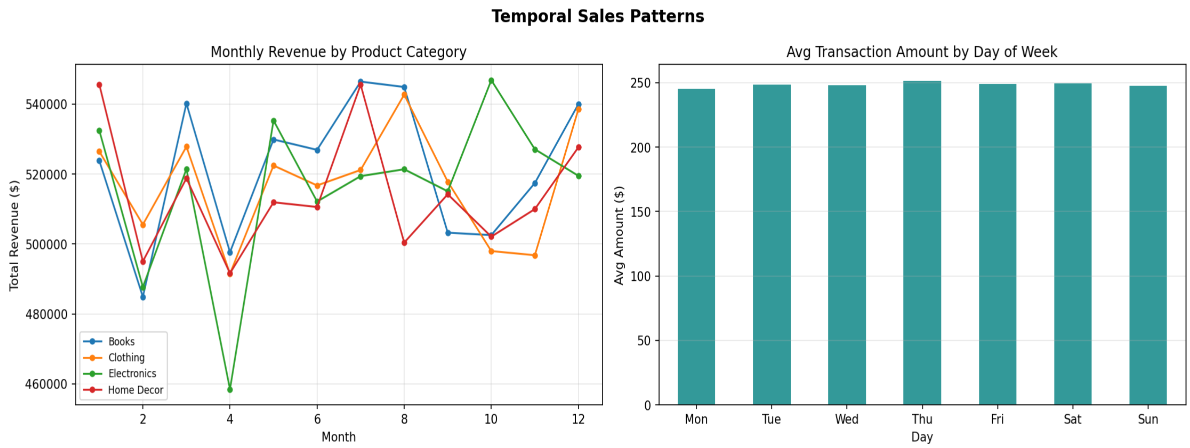

Figure 7. Monthly revenue by category and average transaction amount by day of week.

Monthly revenue is stable across all categories with no significant seasonality (range: $1.94M–$2.13M). Day-of-week averages show minimal variation (Mon $245 – Thu $251), indicating that shopping behavior is not meaningfully influenced by the day of the week. These flat temporal patterns suggest either that the dataset was synthetically generated or that the retail context operates with consistent demand across the calendar.

4. Progress on Discovery Questions

Discovery Question	Status	Key Finding
Which products are frequently purchased together?	ANSWERED (Unexpected)	Category loyalty dominates—94.6% of customers buy one category only. Cross-category lift < 1 for all pairs.
Are there distinct customer segments by purchasing behavior?	ANSWERED	3 segments: Budget Single-Purchase ($142), Repeat Mid-Tier ($503), Premium Single-Purchase ($481).
Which transactions are unusual or anomalous?	ANSWERED	5.0% flagged as anomalous—high amount, high qty, high price, uniform across categories.

5. M4 Plan

M4 will build on M3 findings by adding:

- **Stability testing**: Bootstrap resampling on cluster assignments to confirm k=3 robustness
- **Alternative explanations**: Test whether flat temporal/category patterns are artifacts of synthetic data generation
- **Ethical reflection**: Discuss implications of customer profiling and anomaly flagging on privacy
- **Polished visualizations**: Publication-ready figures with consistent styling
- **Full critical assessment**: Validity, limitations, generalizability of all findings